

Some subgroups of reproductive age women in the United States may be at risk for iodine deficiency.

Consuming an adequate amount of iodine during pregnancy is critical for fetal neurologic development. Even a mild deficiency can impair cognitive ability. Important sources of iodine in the United States include dairy products and iodized salt. Although the U.S. population has traditionally been considered iodine sufficient, median urinary iodine concentrations (UIC) have decreased 50% since the 1970s. We analyzed 2001-2006 NHANES data from urine iodine spot tests for pregnant (n = 326), lactating (n = 53), and nonpregnant, nonlactating (n = 1437) women of reproductive age (15-44 y). We used WHO criteria to define iodine sufficiency (median UIC: 150-249 microg/L among pregnant women; ≥ 100 microg/L among lactating women; and 100-199 microg/L among nonpregnant, nonlactating women). The iodine status of pregnant women was borderline sufficient (median UIC = 153 microg/L; 95% CI = 105-196), while lactating (115 microg/L; 95% CI = 62-162) and nonpregnant, nonlactating (130 microg/L; 95% CI = 117-140) women were iodine sufficient. Dairy product consumption was an important contributor to iodine status among both pregnant and nonpregnant, nonlactating women, and those who do not consume dairy products may be at risk for iodine deficiency. Although larger samples are needed to confirm these findings, these results raise concerns about the iodine status of pregnant women and women of reproductive age who are not consuming dairy products. Iodine levels among U.S. women should be monitored, particularly among subgroups at risk for iodine deficiency.